

Pertinent information

components used				
S.no	Electrical	mechanical	Objectives of the robot	video link
1	Arduino nano/ esp32 microcontroller	palm plate(3D printed PLA)	control the whole hand and provide a rigid contact surface for the high five	.https://youtu.be/5z5evInThP4?si=ta36-jpx29isiF76
2	servo motors (MG90S / MG996R)	Finger phalanges (3 links per figure)	Move the fingers smoothly into open and closed poses	https://www.youtube.com/watch?v=mnurLBNWldc
3	PCA9685 servo driver board	jointa and hinge pins	Drive all the servo from one board and allow low friction finger bending	https://youtu.be/GDnmAl_7lOk?si=YJAtVvygq5UtFtgd
4	ultrasonic (HC-SR04) or IR proximity sensor	Tendlons (fishing line/ nylon cord)	Detect the approaching human hand and transmit servo pull to the fingers	https://youtu.be/A1tPvP2v-ow?si=5uuh6LuPtrXEqEKO
5	Force sensing resistor (FSR)	return springs/ elastic bands	Detect the palm contact at the moment of the high-five and straighten the fingers after release	https://www.youtube.com/results?search_query=Arduino+FSR+force+sensing+resistor+tutorial
6	Flex sensor/ potentiometer	Servo horn and pulley	give free positon feedback and convert servo rotation into tendon pull	https://youtu.be/b7zT94WV-Ek?si=acT-qivlPq_ubDd3
7	LiPo/ 18650 battery pack	Thumb assembly	Supply cordless power and give a realastic hand shape with side support	.https://youtu.be/QOyghUxLdqE?si=2sdoziHsz2YpsZhn
8	Buck converter/ voltage regulator (LM2596)	Servo mounting bracket and forearm frame	provide stable 5-6v to servos and hold them firmly aligned	https://www.youtube.com/results?search_query=Arduino+LM2596+buck+converter+tutorial
9	power switch and fuse protection	wrist/arm mount	Allow safe power control and swing in hand up to meet the incoming palm	https://youtu.be/uEd2B7fS8Eg
10	writing,connectors /PCBV	Fastens and silicone/fore palm pad	make reliable connections and cushion the impact for a safe,quiet high five	https://youtu.be/5z5evInThP4?si=vqMYuEKl_upQrVel